

ADAPTIVE NAVIGATION SYSTEM FOR LINE-FOLLOWING ROBOT

Driftless

Bright Electronics Manufacturing Challenge

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1.0 OBJECTIVES

The objective of this project was to develop an added feature that significantly improves the navigation performance of the line-following robot while complying with the competition requirements. The added feature interfaces only with the microcontroller and supplements the existing line-following algorithm with additional motion information.

The proposed Adaptive Navigation System combines a six-axis inertial measurement unit (IMU) and dual optical wheel encoders to provide continuous measurements of the robot's angular velocity, acceleration, wheel speed, and odometry. These measurements enable predictive navigation, adaptive speed control, and dynamic PID tuning, allowing the robot to navigate corners more efficiently while maintaining accurate line tracking.

2.0 BACKGROUND

Line-following robots are widely used in educational robotics because they provide a straightforward platform for studying autonomous navigation and closed-loop control. Most line-following robots rely almost exclusively on an array of infrared sensors to detect the position of a line and continuously adjust their steering to remain on the course. While this reactive approach is effective for basic navigation, it limits the robot's awareness of its own motion. The microcontroller responds only to what the infrared sensors observe and has no direct knowledge of the robot's speed, angular velocity, acceleration, or wheel movement. As robot speed increases, these limitations become more significant, often resulting in overshoot, oscillation, wheel slip, and reduced performance when navigating corners.

The Bright Electronics Manufacturing Challenge introduced an opportunity to improve the robot by incorporating an added feature. However, the competition rules required that the original robot circuitry remain completely unchanged. Any added functionality therefore had to operate as an independent subsystem that interfaces only with the existing microcontroller. This requirement eliminated the possibility of modifying the original sensor or motor control circuitry and required the added feature to supplement, rather than replace, the existing navigation system.

As students competing in this challenge, we were first interested due to the subject matter. Previously we designed a line following car and the logic behind it in an educational environment, however the car was pre-made, and only basic logic could be changed. This challenge takes it to the next level, allowing us to take full ownership of the design. We also all come from a competitive robotics background, competing in VEX U Robotics and regularly working with autonomous control, sensor integration and odometry. Throughout the design process, we have drawn from our previous experience and knowledge.

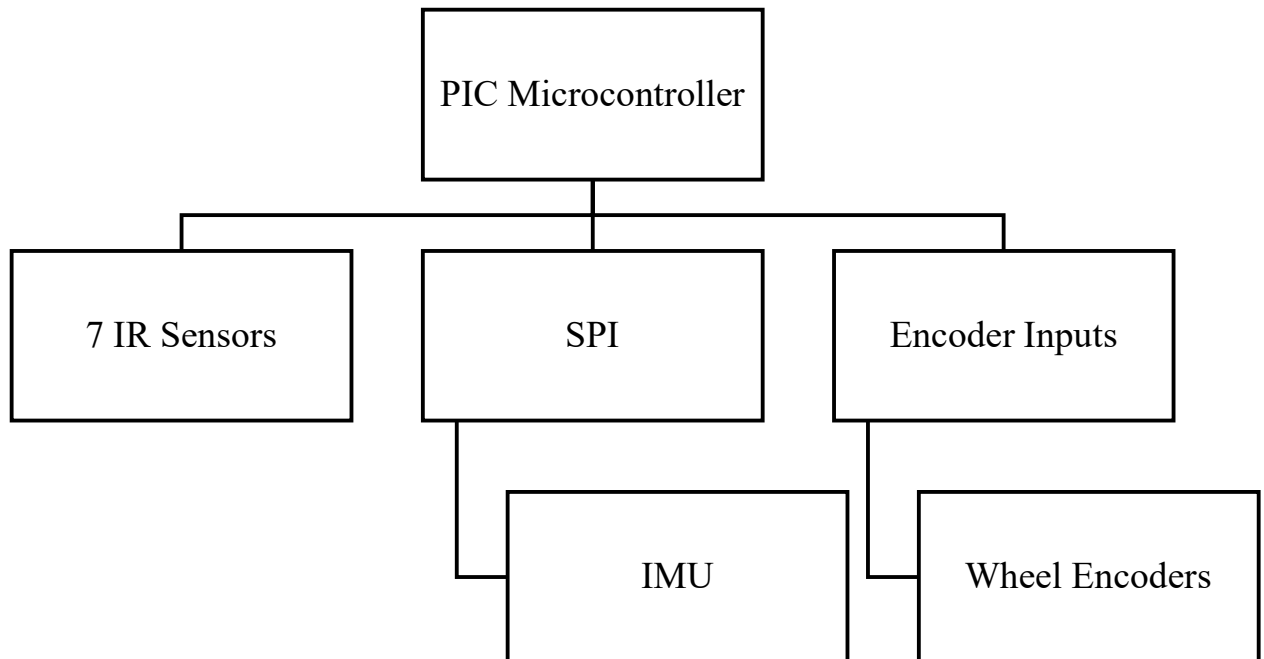
To address these limitations while remaining within the competition constraints, this project proposes an Adaptive Navigation System that combines inertial sensing and wheel

odometry with the existing infrared sensor array. By providing additional real-time measurements of acceleration, angular velocity, wheel speed, and distance traveled, the controller can estimate the robot's motion more accurately and make predictive navigation decisions instead of relying solely on reactive corrections. This approach is expected to improve tracking accuracy, increase cornering performance, and reduce overall lap time while maintaining a compact, low-power implementation.

3.0 DESIGN DESCRIPTION

3.1 System Overview

The Adaptive Navigation System is designed as an independent hardware module that interfaces directly with the existing microcontroller without modifying any portion of the original robot circuitry. The original infrared sensor array continues to perform line detection, while the added IMU and wheel encoders provide additional motion information used by the navigation software.



3.2 IMU

To provide the robot with awareness of its own motion, an inertial measurement unit (IMU) was incorporated into the added feature. Unlike the existing infrared sensors, which only detect the position of the line beneath the robot, the IMU continuously measures the robot's linear acceleration and angular velocity. This additional information allows the microcontroller

to estimate how the robot is moving through turns, how quickly it is rotating, and whether it is accelerating or decelerating. By supplementing the infrared sensor data with inertial measurements, the controller can make more informed navigation decisions rather than relying solely on the line position.

The LSM6DSLTR six-axis IMU was selected because of its low power consumption, compact package, and excellent performance in robotic applications. A six-axis IMU was chosen over a nine-axis device because the additional three-axis magnetometer was unnecessary for a short-duration line-following competition. Eliminating the magnetometer reduced both power consumption and software complexity while still providing all of the motion information required for adaptive navigation. The IMU operates from the robot's 3.3 V supply and consumes approximately 0.65 mA during normal operation, making it well suited for a battery-powered system where minimizing power consumption was a key design objective.

Communication between the IMU and the PIC18F47Q43 microcontroller is accomplished through the Serial Peripheral Interface (SPI). SPI was selected instead of I²C because it provides higher communication bandwidth and full-duplex communication, allowing sensor data to be transferred more quickly and reliably. These advantages are particularly important when sampling the IMU at 208 Hz, where rapid updates of acceleration and angular velocity improve the responsiveness of the adaptive navigation algorithm. The SPI interface consists of the serial clock (SCK), master-out slave-in (MOSI), master-in slave-out (MISO), and chip-select (CS) signals, all of which are routed directly between the IMU and the microcontroller. Unused pins were configured according to the manufacturer's recommendations and either connected to ground or designated as no-connects, as seen in fig.1.

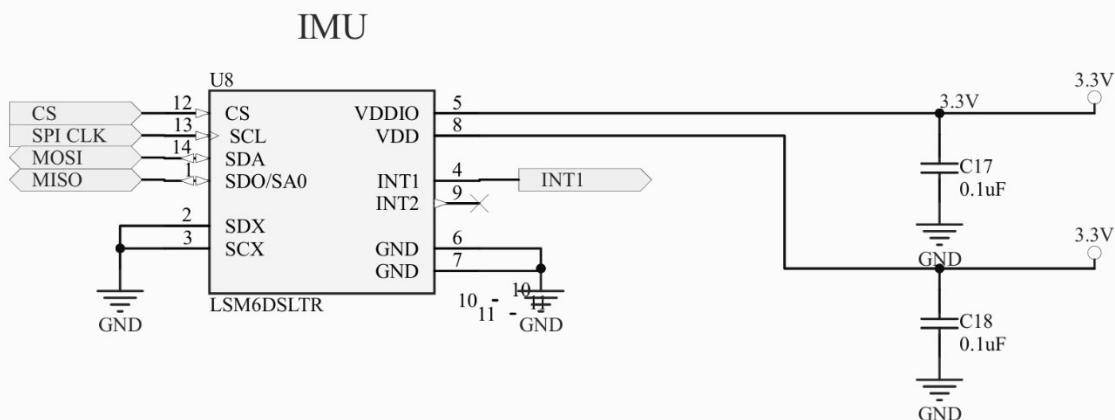


Fig. 1: IMU Schematic

Several PCB design practices were implemented to maximize measurement accuracy and minimize electrical noise. The IMU was intentionally positioned as close as possible to the geometric center of the robot, reducing errors caused by rotational acceleration during cornering and allowing the measured angular velocity to more accurately represent the motion of the entire

vehicle. The device was also placed away from the DC motors and other high-current circuitry to reduce electromagnetic interference generated by motor switching. A continuous ground plane was maintained directly beneath the IMU to provide a low-impedance reference and improve signal integrity.

Power integrity was also carefully considered throughout the design. Bypass capacitors were placed immediately adjacent to the IMU power pins to suppress supply voltage ripple and filter high-frequency noise before it reached the sensor. Similarly, the SPI communication traces were routed as short and direct as possible to minimize propagation delay, crosstalk, and electromagnetic interference. Together, these design decisions improve measurement stability and ensure that the motion data supplied to the adaptive navigation algorithm remains accurate and reliable during high-speed operation.

3.3 Wheel Encoders

To provide the robot with additional information regarding its speed and odometry, which cannot be directly measured by the IMU, dual optical wheel encoders were incorporated into the added feature. While the IMU measures the robot's linear acceleration and angular velocity, it cannot directly determine the rotational speed of the drive wheels. By measuring the actual wheel rotation, the encoders provide an independent measurement of vehicle speed and distance traveled, allowing the controller to compare the commanded motion with the robot's actual movement. This information improves motion estimation and forms the basis for future adaptive speed control, wheel-slip detection, and odometry calculations.

The wheel encoder system consists of OPB360L11 slotted optical switches, as seen in fig. 2, paired with encoder disks, as seen in figure 3, mounted directly to the shafts of the TT motors. The encoder disks contain evenly spaced slots around their circumference. As the wheel rotates, these slots repeatedly interrupt the infrared beam between the emitter and phototransistor within the optical switch. Each interruption produces a digital pulse, with the pulse frequency directly proportional to the rotational speed of the motor. By counting these pulses over time, the microcontroller can accurately determine wheel speed and estimate the distance traveled by each wheel.



Fig. 2: OPB360L11 slotted optical switches

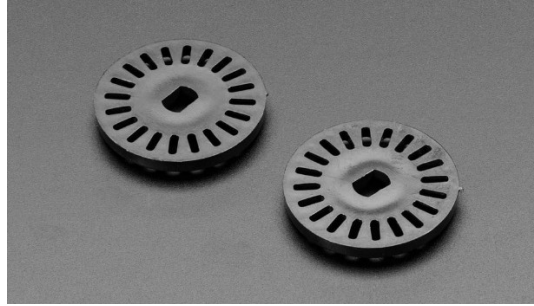


Fig. 3: Wheel encoder disks

Each optical encoder includes an infrared LED emitter that continuously illuminates the encoder disk. A current-limiting resistor was selected to establish the proper forward current through the emitter while minimizing power consumption. On the receiving side, the phototransistor converts the interrupted infrared light into a varying electrical signal corresponding to the passing encoder slots. Because the raw output of the phototransistor can contain slow voltage transitions and electrical noise caused by motor switching and vibration, additional signal conditioning was incorporated before the signals reached the microcontroller.

To ensure reliable operation, the output from each phototransistor is routed through a SN74HC14 Schmitt-trigger inverter. The Schmitt trigger provides hysteresis, eliminating false transitions caused by noise or slow signal edges while producing clean, fast digital pulses suitable for the microcontroller's interrupt inputs. This significantly improves encoder reliability, particularly in the electrically noisy environment created by the DC motors. The inverter also ensures that each encoder pulse has well-defined logic levels, allowing accurate pulse counting even at higher wheel speeds.

Indicator LEDs were also included on each encoder channel to provide immediate visual confirmation that pulses are being generated during operation. These LEDs greatly simplify hardware testing and troubleshooting by allowing the encoder outputs to be verified without requiring diagnostic software or measurement equipment. During development, the LEDs provide a quick indication that the optical sensors are correctly aligned with the encoder disks and that the circuit is functioning properly. The entire wheel encoder schematic can be seen in fig. 4.

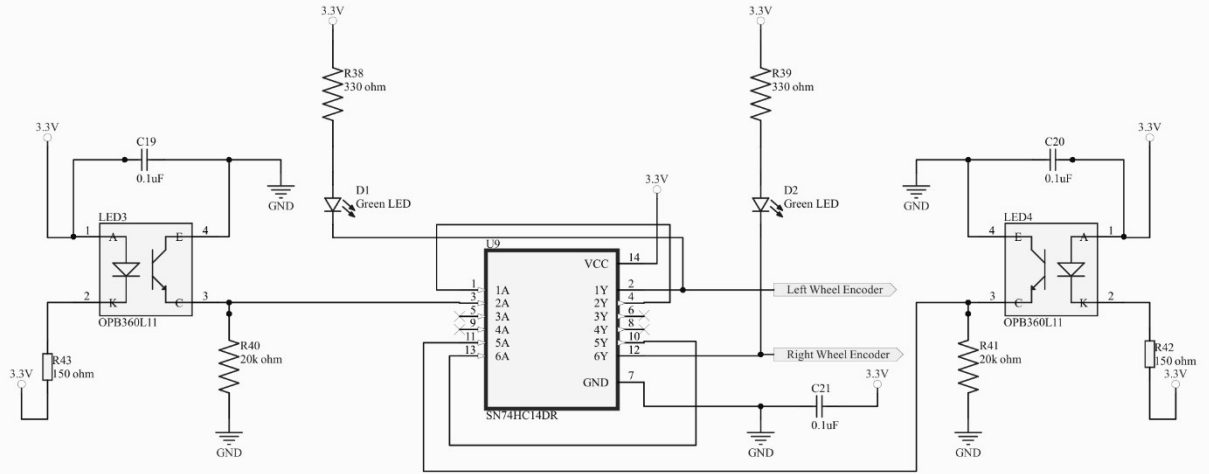


Fig. 4: Wheel Encoder Schematic

The wheel encoder circuitry was positioned close to the mounting locations of the TT motors to minimize trace lengths between the optical sensors and the PCB, reducing susceptibility to electrical noise. The digital outputs from the Schmitt-trigger inverters were then routed directly to dedicated microcontroller pins. This arrangement provides accurate wheel speed measurements while maintaining a compact PCB layout and minimizing routing complexity.

3.4 Engineering Tradeoffs

Throughout the design process, numerous engineering tradeoffs were evaluated to maximize the robot's navigation performance while remaining within the competition constraints of PCB area, power consumption, implementation complexity, and available time. Every component was selected only after considering its expected performance improvement relative to its impact on the overall system.

One of the earliest design decisions was selecting a six-axis IMU instead of a nine-axis device. Although a nine-axis IMU provides an additional three-axis magnetometer for absolute heading measurements, this capability was unnecessary for a short-duration indoor line-following robot. The robot only required measurements of linear acceleration and angular velocity to estimate its motion through corners. Eliminating the magnetometer reduced both power consumption and software complexity while maintaining all of the capability required for the adaptive navigation algorithm.

Communication between the IMU and the microcontroller was implemented using SPI rather than I²C. SPI was selected because it provides higher communication bandwidth, deterministic timing, and full-duplex communication, allowing the IMU to be sampled rapidly. These characteristics improve the responsiveness of the adaptive navigation system, particularly when operating at higher speeds where rapid updates of acceleration and angular velocity are needed.

Several types of wheel encoders were also evaluated before selecting optical slotted encoders. Optical encoders were chosen because they are compact, lightweight, and compatible with the TT motors used on the robot. Their adjustable mounting configuration simplified mechanical integration while still providing sufficient resolution for accurate wheel speed measurement and odometry without significantly increasing the size of the PCB or overall system complexity.

Additional sensing features, including an ambient light sensor and battery monitoring circuitry, were originally considered as part of the Adaptive Navigation System. Although these features could provide useful environmental information, their expected improvement to navigation performance did not justify the additional power consumption, PCB area, routing complexity, and software development time required for implementation. Instead, development effort was focused on integrating the IMU and wheel encoders, which were expected to provide the greatest improvement in robot performance.

Overall, the final design represents a balance between performance, power efficiency, hardware complexity, and implementation risk. Rather than maximizing the number of added sensors, the design focuses on integrating a small number of sensors that provide the greatest improvement in motion estimation and predictive navigation while maintaining a compact, low-power, and reliable system.

3.5 Software Architecture

The Adaptive Navigation System is designed to supplement the existing line-following algorithm rather than replace it. The original infrared sensor array is still responsible for determining the position of the line, while the added IMU and wheel encoders provide additional information regarding the robot's motion. By combining these independent sources of information, the controller develops a more complete understanding of the robot's current operating state than would be possible with only the infrared sensors.

During operation, the microcontroller continuously acquires measurements from the IMU, wheel encoders, and infrared sensor array. The IMU provides real-time measurements of linear acceleration and angular velocity, allowing the controller to determine how quickly the robot is accelerating, decelerating, and rotating as it travels around the course. Simultaneously, the optical wheel encoders measure the rotational speed of each drive wheel and estimate the distance traveled through odometry. These measurements are combined with the infrared sensor data to estimate the robot's current motion state, including its speed, heading, turning rate, and relative position on the line.

The infrared sensor array continues to provide the primary navigation reference by determining the location of the line beneath the robot. Rather than using the sensor outputs only to calculate a tracking error, the software analyzes the response of all seven infrared sensors simultaneously to estimate the geometry of the upcoming section of the track. The distribution of

the active sensors provides an indication of whether the robot is approaching a straight section, a gradual curve, or a sharp corner. This information, combined with the robot's current speed and angular velocity, allows the controller to estimate the severity of the upcoming turn before it is fully entered.

Based on the estimated corner severity, the software predicts the appropriate target speed for the robot before entering the corner rather than reacting only after the line position begins to change. By gradually reducing speed prior to sharp turns and accelerating earlier when exiting corners, the controller is expected to reduce overshoot while maintaining the highest possible average speed around the course. This predictive approach allows the robot to remain more stable during aggressive maneuvers and reduces unnecessary braking on gradual turns.

An adaptive PID controller is then used to adjust the steering response according to the current operating conditions. Rather than using a single fixed set of controller gains, the proportional, integral, and derivative gains are continuously adjusted based on factors such as robot speed and estimated corner severity. During straight sections, higher gains and increased target speeds allow the robot to maximize performance. As corner severity increases, the controller gradually reduces the target speed while modifying the PID gains to produce smoother steering corrections, minimizing oscillation and overshoot. This adaptive gain scheduling allows the robot to remain responsive over a wider range of operating conditions than would be possible with a conventional fixed-gain controller.

Although the adaptive navigation software builds upon the existing line-following algorithm, the additional motion information provided by the IMU and wheel encoders allows the controller to make predictive navigation decisions instead of relying solely on reactive corrections. This combination of inertial sensing, wheel odometry, and infrared line detection provides a more robust estimate of the robot's motion and is expected to improve both tracking accuracy and overall lap time.

4.0 TECHNICAL SPECIFICATIONS

The Adaptive Navigation System was designed to provide the maximum navigation improvement while minimizing power consumption, PCB area, and implementation complexity. Table 1 summarizes the primary hardware specifications of the added feature.

Table 1: Hardware Specifications

Item	Specification
IMU	LSM6DSLTR
Supply Voltage	3.3V
Microcontroller Interface	SPI
Sample Rate	208 Hz
Wheel Encoders	Dual OPB360L11
Wheel Encoder Logic	SN74HC14 Schmitt Trigger
PCB Stack-up	4 Layer

Communication	SPI
Added Power	163 mW
Sensors Added	3

The LSM6DSLTR was selected because it provides six-axis inertial sensing with very low power consumption, making it well suited for battery-powered robotic applications. Operating from the robot's regulated 3.3 V supply simplifies power distribution while remaining compatible with the existing microcontroller. The IMU is sampled at 208 Hz, providing frequent motion updates that allow the adaptive controller to respond quickly to changes in robot motion without placing excessive processing demands on the microcontroller.

Communication with the PIC18F47Q43 is performed using SPI) SPI was selected instead of I²C because it provides higher communication bandwidth, deterministic timing, and full-duplex data transfer, allowing the IMU measurements to be updated with minimal latency during high-speed operation.

Robot speed and odometry are measured using dual OPB360L11 optical wheel encoders, with each encoder output conditioned by an SN74HC14 Schmitt-trigger inverter before being processed by the microcontroller. The Schmitt trigger converts the analog phototransistor output into clean digital pulses, improving noise immunity and ensuring reliable pulse counting even in the electrically noisy environment created by the drive motors.

The added circuitry is implemented on a four-layer printed circuit board consisting of dedicated power and ground planes with top and bottom signal routing layers. Additionally, ground pours were added on the top and bottom signal layers. This stack-up improves signal integrity, reduces electromagnetic interference, and provides a compact implementation while maintaining reliable operation of the high-speed SPI interface and encoder circuitry.

4.1 Power Consumption

Power consumption was considered throughout the design process because the robot operates from a limited battery supply. The final implementation consumes approximately 163 mW, allowing the Adaptive Navigation System to provide significant additional sensing capability while contributing only a small increase to the overall power budget.

POWER CALCULATIONS				
<i>Component</i>	<i>Current (Typical)</i>	<i>Power</i>	<i>Quantity</i>	<i>Total</i>
LSM6DSLTR (IMU)	0.65 mA	2.145 mW	1	2.145 mW
OPB360L11 (LED)	14 mA	46.2 mW	2	92.4 mW
OPB360L11 (Phototransistor)	10 mA	33 mW	2	66 mW
150060VS75000 (Green LED)	0.1 mA	0.33 mW	2	0.66 mW
SN74HC14DR (Schmitt Trigger Inverter)	0.4 mA	1.32 mW	1	1.32 mW
Total Added Power				162.53 mW

Although the infrared emitters within the optical wheel encoders account for most of the added power consumption, they provide the high-resolution wheel speed measurements necessary for odometry and adaptive navigation. In comparison, the IMU contributes only 2.15 mW, demonstrating that advanced inertial sensing can be incorporated with minimal impact on battery life. Overall, the final design achieves a balance between navigation performance, sensing capability, and power efficiency, meeting the project's objective of significantly improving robot performance while maintaining a low-power implementation.

5.0 IMPLEMENTATION STRATEGY

5.1 Hardware

Implementation begins with assembling and validating the Adaptive Navigation System independently of the base robot using a digital multimeter. The four-layer PCB is populated and each subsystem is tested individually before being integrated with the robot. Power rails are verified first, followed by SPI communication with the IMU and digital signal verification of the wheel encoders. The encoder indicator LEDs provide immediate visual confirmation that encoder pulses are being generated correctly before software development begins. Once each subsystem is operating correctly, the board is connected to the robot through the designated microcontroller interface while leaving the original robot circuitry unmodified.

5.2 Software

Software development follows a modular approach. The IMU and wheel encoder drivers are first developed independently to verify reliable sensor communication and accurate data acquisition. Next, the encoder pulse counting and odometry algorithms are implemented, followed by integration of the IMU measurements with the existing infrared sensor array. Finally, the adaptive navigation algorithms are developed, including corner severity estimation, predictive speed planning, and adaptive PID gain scheduling. Each software module is verified individually before being integrated into the complete navigation system.

5.3 Testing and Validation

System validation is performed in stages. Individual hardware components are first tested to verify proper operation and communication. Sensor outputs are then monitored while the robot is driven manually to validate acceleration, angular velocity, and wheel speed measurements. After successful hardware verification, the adaptive navigation algorithms are tuned on progressively more challenging line-following courses. Performance will then be evaluated through:

- Average lap time

- Maximum stable corner speed
- Line tracking error
- Overshoot entering corners
- Number of line losses
- Wheel encoder accuracy
- IMU communication accuracy

5.4 Design Risks

Several technical risks were identified during development. Electrical noise generated by the DC motors could interfere with both the IMU measurements and the optical encoder signals. This risk was mitigated through careful PCB layout, short SPI trace routing, continuous ground planes, bypass capacitors, and Schmitt-trigger signal conditioning.

Mechanical alignment of the optical wheel encoders also presents a potential challenge. Improper alignment between the encoder disk and optical switch may result in missed encoder pulses or inconsistent speed measurements. 3D models of the TT motors and wheel encoders were created to ensure the distance is sufficient.

Finally, because the adaptive navigation software relies on multiple sensor inputs, additional tuning is required to determine the optimal PID gains and corner prediction parameters. This risk is addressed through incremental testing and validation of each subsystem before complete integration.

6.0 CONCLUSION / RECOMMENDATIONS

The proposed Adaptive Navigation System provides a practical method of significantly improving the performance of the competition line-following robot while fully complying with the competition requirement that the original robot circuitry remain unchanged. By integrating a low-power six-axis IMU with dual optical wheel encoders, the robot gains continuous measurements of acceleration, angular velocity, wheel speed, and odometry that are unavailable using the existing infrared sensors alone.

These additional measurements enable predictive corner detection, adaptive speed planning, and dynamic PID gain scheduling, allowing the controller to make more informed navigation decisions rather than relying solely on reactive line tracking. Careful component selection, low-power design practices, and a four-layer PCB layout ensure that these improvements are achieved with minimal impact on the robot's overall power budget and board space. The proposed design provides a strong foundation for future expansion while meeting all competition requirements for the added feature challenge.